

Baseline

For our initial quantitative baseline, we used a Logistic Regression classifier trained on text features to predict sentiment labels (positive, negative, neutral) from the drug reviews dataset. We evaluated the model using both accuracy and macro-F1 scores.

Our baseline static model (TF-IDF) achieved an overall accuracy of **0.780** and a macro-F1 score of **0.532**. We chose macro-F1 as our primary evaluation metric because our dataset is heavily imbalanced. Looking at the data split, there are 33,150 positive reviews compared to only 4,439 neutral ones. If we relied solely on accuracy, a model that guesses "positive" every time would score high while completely failing to find neutral or negative sentiments. The macro-F1 score treats all classes equally, exposing the fact that our baseline severely struggles with the minority class (scoring a dismal 0.07 F1-score on neutral).

Static vs. Contextual

Switching from a static TF-IDF representation to contextual MiniLM embeddings yielded an unexpected result: performance actually dropped. The MiniLM model scored an accuracy of **0.749** and a macro-F1 of **0.480**.

TF-IDF builds a static representation based on vocabulary frequency. It works well here because drug reviews often contain highly predictive, explicit sentiment words that map directly to the target label regardless of context. MiniLM, on the other hand, creates dense, contextual embeddings where the representation of a word changes depending on its surroundings. While contextual models are generally superior for complex linguistic structures, our specific dense model (all-MiniLM-L6-v2) compressed the text into 384 dimensions without explicit fine-tuning for sentiment. For this specific classification task, the raw frequency of explicit keywords in the sparse 5,000-dimension TF-IDF matrix provided a stronger signal to the linear classifier than the generalized semantic space of the pre-trained embeddings.

Bias

We performed a quick bias check by evaluating the accuracy of the MiniLM model across two distinct categories: mental_health conditions (like ADHD, Depression, and Anxiety) versus all other medical conditions.

The group check revealed a slight performance gap: the model achieved a **0.649** accuracy on the mental_health cohort compared to **0.603** on other conditions. While a 4.6% gap is noticeable, it shows the model actually performs slightly better on mental health reviews. If we discovered a significant negative gap against a vulnerable group, there would be a fairness issue and would suggest the embedding space or the classifier fails to capture the unique vocabulary and nuances of that group's experiences. To mitigate the gap, we would look into balancing our training data across groups, applying class-weight adjustments during training, or exploring domain-specific embeddings.

Next Steps

To beat the baseline, we may want to move away from simple linear classifiers. We will explore

fine-tuning a transformer model (like BERT or DistilBERT) directly on our dataset so the embeddings learn the specific contextual nuances of pharmaceutical reviews. Additionally, we want to implement text-specific preprocessing to handle contractions and artifacts better, and experiment with advanced techniques for handling class imbalance, such as SMOTE or adjusting the loss function to penalize minority class errors more heavily. Finally, we would need to test our models with the full dataset of over 161k reviews, as for this exercise we only worked on a sample of 50,000.